

Chapter 2 IS-IS Features and Configurations

Lab 2-1 IS-IS Configurations

Learning Objectives

The objectives of this lab are to learn and understand:

- Basic IS-IS configurations
- How to configure the IS-IS designated intermediate system (DIS) priority
- How to configure the IS-IS network type
- How to import external routes into IS-IS
- How to configure the IS-IS interface cost
- How to configure IS-IS route leaking

Topology

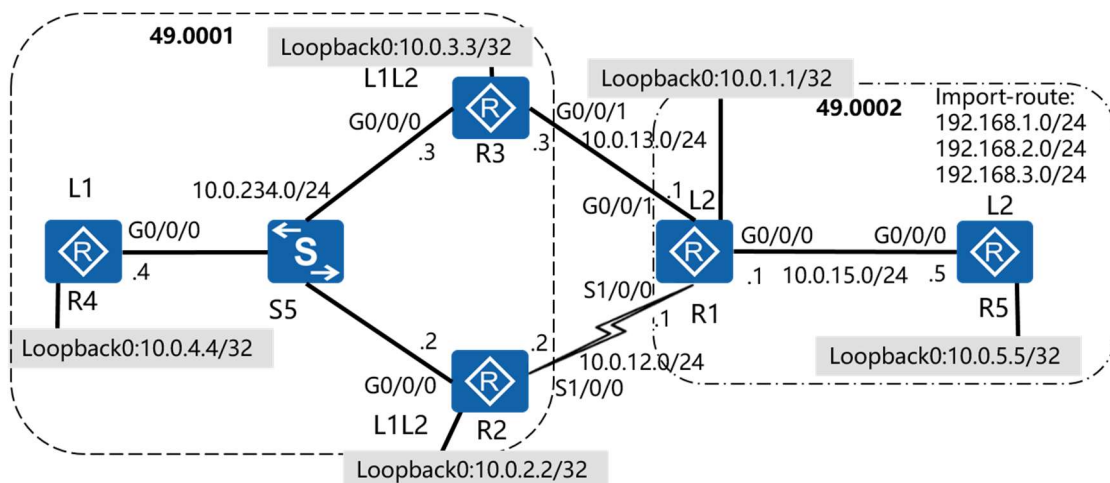


Figure 3-1 IS-IS topology

Scenario

IS-IS runs as an IGP in a network. R1 and R5 run in Area 49.0002 as Level-2 routers. R2, R3, and R4 run in Area 49.0001. R4 is a Level-1 router, while R2 and R3 are Level-1-2 routers. The requirements are as follows: R4 can use the external routes

imported by R5. R4 connected to S5 functions as the DIS. There is a P2P link between R1 and R5. Uplink and downlink traffic from R4 to R5 is forwarded through Ethernet interfaces, and route selection is controlled using the cost and route leaking. Switches do not require additional configurations and are only responsible for transparent forwarding.

Tasks

Step 1 **Set basic parameters and configure IP addresses.**

Configure IP addresses for all the routers.

```
[R1]interface loopback 0
[R1-LoopBack0]ip address 10.0.1.1 32
[R1-LoopBack0]quit
[R1]interface GigabitEthernet 0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.0.15.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]interface GigabitEthernet 0/0/1
[R1-GigabitEthernet0/0/1]ip address 10.0.13.1 24
[R1-GigabitEthernet0/0/1]quit
[R1]interface interface Serial1/0/0
[R1-Serial1/0/0]ip address 10.0.12.1 24
[R1-Serial1/0/0]quit
```

```
[R2]interface LoopBack 0
[R2-LoopBack0]ip address 10.0.2.2 32
[R2-LoopBack0]quit
[R2]interface GigabitEthernet 0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.0.234.2 24
[R2-GigabitEthernet0/0/0]quit
[R2]interface Serial1/0/0
```

```
[R2-Serial1/0/0]ip address 10.0.12.2 24
```

```
[R2-Serial1/0/0]quit
```

```
[R3]interface LoopBack 0
```

```
[R3-LoopBack0]ip address 10.0.3.3 32
```

```
[R3-LoopBack0]quit
```

```
[R3]interface GigabitEthernet 0/0/0
```

```
[R3-GigabitEthernet0/0/0]ip address 10.0.234.3 24
```

```
[R3-GigabitEthernet0/0/0]quit
```

```
[R3]interface GigabitEthernet 0/0/1
```

```
[R3-GigabitEthernet0/0/1]ip address 10.0.13.3 24
```

```
[R3-GigabitEthernet0/0/1]quit
```

```
[R4]interface LoopBack 0
```

```
[R4-LoopBack0]ip address 10.0.4.4 32
```

```
[R4-LoopBack0]quit
```

```
[R4]interface GigabitEthernet 0/0/0
```

```
[R4-GigabitEthernet0/0/0]ip address 10.0.234.4 24
```

```
[R4-GigabitEthernet0/0/0]quit
```

```
[R5]interface LoopBack 0
```

```
[R5-LoopBack0]ip address 10.0.5.5 32
```

```
[R5-LoopBack0]quit
```

```
[R5]interface GigabitEthernet 0/0/0
```

```
[R5-GigabitEthernet0/0/0]ip address 10.0.15.5 24
```

```
[R5-GigabitEthernet0/0/0]quit
```

After the configurations are complete, test connectivity from R1 to directly connected interfaces of R2, R3, and R5.

```
[R1]ping -c 1 10.0.13.3
```

```
PING 10.0.13.3: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.13.3: bytes=56 Sequence=1 ttl=255 time=1 ms
```

```
--- 10.0.13.3 ping statistics ---
```

```
1 packet(s) transmitted
```

```
1 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 1/1/1 ms
```

```
[R1]ping -c 1 10.0.12.2
```

```
PING 10.0.12.2: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.12.2: bytes=56 Sequence=1 ttl=255 time=26 ms
```

```
--- 10.0.12.2 ping statistics ---
```

```
1 packet(s) transmitted
```

```
1 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 26/26/26 ms
```

```
[R1]ping -c 1 10.0.15.5
```

```
PING 10.0.15.5: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.15.5: bytes=56 Sequence=1 ttl=255 time=1 ms
```

```
--- 10.0.15.5 ping statistics ---
```

```
1 packet(s) transmitted
```

```
1 packet(s) received
```

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms

Test connectivity from R4 to directly connected interfaces of R2 and R3.

[R4]ping -c 1 10.0.234.2

PING 10.0.234.2: 56 data bytes, press CTRL_C to break

Reply from 10.0.234.2: bytes=56 Sequence=1 ttl=255 time=1 ms

--- 10.0.234.2 ping statistics ---

1 packet(s) transmitted

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms

[R4]ping -c 1 10.0.234.3

PING 10.0.234.3: 56 data bytes, press CTRL_C to break

Reply from 10.0.234.3: bytes=56 Sequence=1 ttl=255 time=1 ms

--- 10.0.234.3 ping statistics ---

1 packet(s) transmitted

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms

Step 2 **Perform basic IS-IS configurations.**

Configure an IS-IS process 1 for each router according to the topology. The following example configures IS-IS process 1 for R1.

R1 resides in Area 49.0002 and uses network-entity 49.0002.0000.0000.0001.

```
[R1]isis
```

```
[R1-isis-1]network-entity 49.0002.0000.0000.0001.00
```

By default, after an IS-IS process is enabled on a router, the router works in Level-1-2 mode. According to the planning, R1 needs to work in Level-2 mode. Therefore, you need to change its IS level.

```
[R1-isis-1]is-level level-2
```

```
[R1-isis-1]quit
```

Enable IS-IS on related interfaces, including loopback interfaces. When no IS-IS process ID is specified, by default, IS-IS is enabled in IS-IS process 1.

```
[R1]interface LoopBack 0
```

```
[R1-LoopBack0]isis enable
```

```
[R1-LoopBack0]quit
```

```
[R1]interface GigabitEthernet 0/0/0
```

```
[R1-GigabitEthernet0/0/0]isis enable
```

```
[R1-GigabitEthernet0/0/0]quit
```

```
[R1]interface GigabitEthernet 0/0/1
```

```
[R1-GigabitEthernet0/0/1]isis enable
```

```
[R1-GigabitEthernet0/0/1]quit
```

```
[R1]interface Serial 1/0/0
```

```
[R1-Serial1/0/0]isis enable
```

```
[R1-Serial1/0/0]quit
```

On R1, check IS-IS enabling information on interfaces.

```
[R1]display isis interface
```

Interface information for ISIS(1)

Interface	Id	IPV4.State	IPV6.State	MTU	Type	DIS
Loop0	001	Up	Down	1500	L1/L2	--
GE0/0/0	001	Up	Down	1497	L1/L2	No/No
GE0/0/1	002	Up	Down	1497	L1/L2	No/No
S1/0/0	002	Up	Down	1500	L1/L2	--

The preceding command output shows that ISIS(1) has been enabled on a total of four interfaces, whose **IPV4.State** field displays Up.

Similarly, configure other routers. R2 and R3 work in Level-1-2 mode, so you do not need to change their IS levels.

```
[R2]isis 1
```

```
[R2-isis-1]network-entity 49.0001.0000.0000.0002.00
```

```
[R2-isis-1]quit
```

```
[R2]interface LoopBack 0
```

```
[R2-LoopBack0]isis enable
```

```
[R2-LoopBack0]quit
```

```
[R2]interface GigabitEthernet 0/0/0
```

```
[R2-GigabitEthernet0/0/0]isis enable
```

```
[R2-GigabitEthernet0/0/0]quit
```

```
[R2]interface Serial 1/0/0
```

```
[R2-Serial1/0/0]isis enable
```

```
[R2-Serial1/0/0]quit
```

On R2, check IS-IS enabling information on interfaces.

```
[R2]display isis interface
```

Interface information for ISIS(1)

Interface	Id	IPv4.State	IPv6.State	MTU	Type	DIS
Loop0	002	Up	Down		1500 L1/L2 --	
GE0/0/0	001	Up	Down		1497 L1/L2 No/No	
S1/0/0	001	Up	Down		1500 L1/L2 --	

Configure IS-IS on R3.

```
[R3]isis 1
[R3-isis-1]network-entity 49.0001.0000.0000.0003.00
[R3-isis-1]quit
[R3]interface LoopBack 0
[R3-LoopBack0]isis enable
[R3-LoopBack0]quit
[R3]interface GigabitEthernet 0/0/0
[R3-GigabitEthernet0/0/0]isis enable
[R3-GigabitEthernet0/0/0]quit
[R3]interface GigabitEthernet 0/0/1
[R3-GigabitEthernet0/0/1]isis enable
[R3-GigabitEthernet0/0/1]quit
```

On R3, check IS-IS enabling information on interfaces.

```
[R3]display isis interface
```

Interface information for ISIS(1)

Interface	Id	IPv4.State	IPv6.State	MTU	Type	DIS
Loop0	001	Up	Down	1500	L1/L2	--
GE0/0/0	002	Up	Down	1497	L1/L2	No/No
GE0/0/1	001	Up	Down	1497	L1/L2	No/No

Configure IS-IS on R4.

```
[R4]isis 1
[R4-isis-1]network-entity 49.0001.0000.0000.0004.00
[R4-isis-1]is-level level-1
[R4-isis-1]quit
[R4]interface LoopBack 0
[R4-LoopBack0]isis enable
[R4-LoopBack0]quit
[R4]interface GigabitEthernet 0/0/0
[R4-GigabitEthernet0/0/0]isis enable
[R4-GigabitEthernet0/0/0]quit
```

On R4, check IS-IS enabling information on interfaces.

```
[R4]display isis interface
```

Interface information for ISIS(1)

Interface	Id	IPv4.State	IPv6.State	MTU	Type	DIS
Loop0	001	Up	Down	1500	L1/L2	--
GE0/0/0	001	Up	Down	1497	L1/L2	No/No

Configure IS-IS on R5.

```
[R5]isis 1
[R5-isis-1]network-entity 49.0002.0000.0000.0005.00
[R5-isis-1]is-level level-2
[R5-isis-1]quit
[R5]interface LoopBack 0
[R5-LoopBack0]isis enable
[R5-LoopBack0]quit
[R5]interface GigabitEthernet 0/0/0
[R5-GigabitEthernet0/0/0]isis enable
[R5-GigabitEthernet0/0/0]quit
```

On R5, check IS-IS enabling information on interfaces.

```
[R5]display isis interface
```

Interface information for ISIS(1)

Interface	Id	IPV4.State	IPV6.State	MTU	Type	DIS
Loop0	001	Up	Down	1500	L1/L2	--
GE0/0/0	001	Up	Down	1497	L1/L2	No/No

After the configurations are complete, check IS-IS neighbor statuses of routers. The following example displays the IS-IS neighbor status of R1. R1 has three neighbors: R2, R3, and R5.

```
[R1]display isis peer
```

Peer information for ISIS(1)

System Id	Interface	Circuit Id	State	HoldTime	Type	PRI
-----------	-----------	------------	-------	----------	------	-----

0000.0000.0005	GE0/0/0	0000.0000.0005.01	Up	7s	L2	64
0000.0000.0003	GE0/0/1	0000.0000.0001.02	Up	21s	L2	64
0000.0000.0002	S1/0/0	0000000001	Up	28s	L2	--

Total Peer(s): 3

In the preceding command output, the **System Id** field is similar to the **Router Id** field of other routing protocols. You can see that R2, R3, and R5 are in Up state. Their IS-IS neighbor relationships with R1 are normal.

Continue to check IS-IS neighbor statuses of other devices.

[R2]display isis peer

Peer information for ISIS(1)

System Id	Interface	Circuit Id	State	HoldTime	Type	PRI
0000.0000.0001	S1/0/0	0000000001	Up	22s	L2	--
0000.0000.0003	GE0/0/0	0000.0000.0004.01	Up	24s	L1(L1L2)	64
0000.0000.0004	GE0/0/0	0000.0000.0004.01	Up	7s	L1	64
0000.0000.0003	GE0/0/0	0000.0000.0002.01	Up	26s	L2(L1L2)	64

Total Peer(s): 4

[R3]display isis peer

Peer information for ISIS(1)

System Id	Interface	Circuit Id	State	HoldTime	Type	PRI

0000.0000.0001	GE0/0/1	0000.0000.0001.02	Up	8s	L2	64
0000.0000.0002	GE0/0/0	0000.0000.0004.01	Up	30s	L1(L1L2)	64
0000.0000.0004	GE0/0/0	0000.0000.0004.01	Up	7s	L1	64
0000.0000.0002	GE0/0/0	0000.0000.0002.01	Up	9s	L2(L1L2)	64

Total Peer(s): 4

[R4]display isis peer

Peer information for ISIS(1)

System Id	Interface	Circuit Id	State	HoldTime	Type	PRI

0000.0000.0003	GE0/0/0	0000.0000.0004.01	Up	29s	L1	64
0000.0000.0002	GE0/0/0	0000.0000.0004.01	Up	23s	L1	64

Total Peer(s): 2

[R5]display isis peer

Peer information for ISIS(1)

System Id	Interface	Circuit Id	State	HoldTime	Type	PRI

0000.0000.0001	GE0/0/0	0000.0000.0005.01	Up	29s	L2	64

Total Peer(s): 1

Step 3 Change the DIS priority.

R2, R3, and R4 establish IS-IS neighbor relationships in a broadcast network. Therefore, the DIS needs to be elected. By default, the DIS priority is 64. If these routers have the same DIS priority, the router with the highest MAC address becomes the DIS. To ensure that R4 becomes the DIS, change its DIS priority.

```
[R4]interface GigabitEthernet 0/0/0
```

```
[R4-GigabitEthernet0/0/0]isis dis-priority 120
```

```
[R4-GigabitEthernet0/0/0]quit
```

```
[R4]display isis interface
```

Interface information for ISIS(1)

Interface	Id	IPv4.State	IPv6.State	MTU	Type	DIS
GE0/0/0	001	Up	Down	1497	L1/L2	Yes/No
Loop0	001	Up	Down	1500	L1/L2	–

Check DIS priorities of neighbors on R2 and R3.

```
[R2]display isis peer
```

Peer information for ISIS(1)

System Id	Interface	Circuit Id	State	HoldTime	Type	PRI
0000.0000.0001	S1/0/0	0000000001	Up	29s	L2	--
0000.0000.0003	GE0/0/0	0000.0000.0004.01	Up	25s	L1(L1L2)	64
0000.0000.0004	GE0/0/0	0000.0000.0004.01	Up	8s	L1	120

```
0000.0000.0003 GE0/0/0          0000.0000.0002.01 Up   20s    L2(L1L2)    64
```

Total Peer(s): 4

```
[R3]display isis peer
```

Peer information for ISIS(1)

System Id	Interface	Circuit Id	State	HoldTime	Type	PRI
0000.0000.0001	GE0/0/1	0000.0000.0001.02	Up	8s	L2	64
0000.0000.0002	GE0/0/0	0000.0000.0004.01	Up	22s	L1(L1L2)	64
0000.0000.0004	GE0/0/0	0000.0000.0004.01	Up	7s	L1	120
0000.0000.0002	GE0/0/0	0000.0000.0002.01	Up	8s	L2(L1L2)	64

Total Peer(s): 4

Step 4 Configure the IS-IS network type.

In a broadcast network, by default, an IS-IS router sets the circuit-type of interfaces to broadcast and participates in DIS election. In the topology, the Ethernet between R1 and R5 has only two routers. You can set the circuit-type of interfaces between the two routers to P2P for optimization.

```
[R1]interface GigabitEthernet 0/0/0
```

```
[R1-GigabitEthernet0/0/0]isis circuit-type p2p
```

```
[R1-GigabitEthernet0/0/0]quit
```

```
[R5]interface GigabitEthernet 0/0/0
```

```
[R5-GigabitEthernet0/0/0]isis circuit-type p2p
```

```
[R5-GigabitEthernet0/0/0]quit
```

During the change of the circuit-type, neighbor relationships are established again. Check the configuration. The **Circuit Id** field format changes and the **Circuit Parameters** field displays **p2p**. The following uses the display of R1 as an example.

```
[R1]display isis peer
```

Peer information for ISIS(1)

System Id	Interface	Circuit Id	State	HoldTime	Type	PRI

0000.0000.0005	GE0/0/0	00000000002	Up	22s	L2	--
0000.0000.0003	GE0/0/1	0000.0000.0001.02	Up	27s	L2	64
0000.0000.0002	S1/0/0	00000000001	Up	22s	L2	--

```
[R1]display isis interface GigabitEthernet 0/0/0 verbose
```

Interface information for ISIS(1)

Interface	Id	IPv4.State	IPv6.State	MTU	Type	DIS
GE0/0/0	003	Up	Down	1497	L1/L2	--
Circuit MT State		: Standard				
Circuit Parameters		: p2p				
Description		: HUAWEI, AR Series, GigabitEthernet0/0/0 Interface				
SNPA Address		: d0d0-4b03-d3fc				
IP Address		: 10.0.15.1				
IPv6 Link Local Address		:				
IPv6 Global Address(es)		:				

```

Csnp Timer Value      : L12    10
Hello Timer Value      :          10
DIS Hello Timer Value  :
Hello Multiplier Value :      3
Cost                   : L1      10 L2      10
Ipv6 Cost              : L1      10 L2      10
Retransmit Timer Value : L12      5
LSP-Throttle Timer     : L12     50
Bandwidth-Value        : Low    100000000 High      0
Static Bfd             : NO
Dynamic Bfd            : NO
Fast-Sense Rpr         : NO
Extended-Circuit-Id Value : 0000000003

```

Step 5 **Configure IS-IS to import external routes.**

Before importing external routes into IS-IS, check current route learning. The following command output shows that traffic from R1 to R4 is load balanced between GE0/0/1 and S1/0/0.

[R1]display isis route

```

Route information for ISIS(1)

```

```

-----

```

```

ISIS(1) Level-2 Forwarding Table

```

```

-----

```

IPV4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags

10.0.4.4/32	20	NULL	GE0/0/1	10.0.13.3	A/-/-/-
			S1/0/0	10.0.12.2	
10.0.5.5/32	10	NULL	GE0/0/0	10.0.15.5	A/-/-/-
10.0.12.0/24	10	NULL	S1/0/0	Direct	D/-/L/-
10.0.13.0/24	10	NULL	GE0/0/1	Direct	D/-/L/-
10.0.234.0/24	20	NULL	S1/0/0	10.0.12.2	A/-/-/-
			GE0/0/1	10.0.13.3	
10.0.15.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.1.1/32	0	NULL	Loop0	Direct	D/-/L/-
10.0.2.2/32	10	NULL	S1/0/0	10.0.12.2	A/-/-/-
10.0.3.3/32	10	NULL	GE0/0/1	10.0.13.3	A/-/-/-

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,
U-Up/Down Bit Set

[R1]display ip routing-table protocol isis

Route Flags: R - relay, D - download to fib

Public routing table : ISIS

Destinations : 5 Routes : 7

ISIS routing table status : <Active>

Destinations : 5 Routes : 7

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.2.2/32	ISIS-L2	15	10	D	10.0.12.2	Serial1/0/0
10.0.3.3/32	ISIS-L2	15	10	D	10.0.13.3	GigabitEthernet0/0/1
10.0.4.4/32	ISIS-L2	15	20	D	10.0.13.3	GigabitEthernet0/0/1
	ISIS-L2	15	20	D	10.0.12.2	Serial1/0/0

10.0.5.5/32	ISIS-L2	15	10	D	10.0.15.5	GigabitEthernet0/0/0
10.0.234.0/24	ISIS-L2	15	20	D	10.0.12.2	Serial1/0/0
	ISIS-L2	15	20	D	10.0.13.3	GigabitEthernet0/0/1

ISIS routing table status : <Inactive>

Destinations : 0 Routes : 0

R2 is a Level-1-2 router and so generates different routes for Level-1 and Level-2 routers. For Level-1 router, it generates a default route pointing to the null interface. This situation also exists on R3.

[R2]display isis route

Route information for ISIS(1)

ISIS(1) Level-1 Forwarding Table

IPv4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags

0.0.0.0/0	10	NULL			
10.0.4.4/32	10	NULL	GE0/0/0	10.0.234.4	A/-/L/-
10.0.12.0/24	10	NULL	S1/0/0	Direct	D/-/L/-
10.0.13.0/24	20	NULL	GE0/0/0	10.0.234.3	A/-/L/-
10.0.234.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.2.2/32	0	NULL	Loop0	Direct	D/-/L/-
10.0.3.3/32	10	NULL	GE0/0/0	10.0.234.3	A/-/L/-

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,

U-Up/Down Bit Set

ISIS(1) Level-2 Forwarding Table

IPv4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags

10.0.4.4/32	20	NULL			
10.0.5.5/32	20	NULL	S1/0/0	10.0.12.1	A/-/-/-
10.0.12.0/24	10	NULL	S1/0/0	Direct	D/-/L/-
10.0.13.0/24	20	NULL			
10.0.234.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.15.0/24	20	NULL	S1/0/0	10.0.12.1	A/-/-/-
10.0.1.1/32	10	NULL	S1/0/0	10.0.12.1	A/-/-/-
10.0.2.2/32	0	NULL	Loop0	Direct	D/-/L/-
10.0.3.3/32	10	NULL			

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,

U-Up/Down Bit Set

[R2]display ip routing-table protocol isis

Route Flags: R - relay, D - download to fib

Public routing table : ISIS

Destinations : 6 Routes : 6

ISIS routing table status : <Active>

Destinations : 6 Routes : 6

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
------------------	-------	-----	------	-------	---------	-----------

10.0.1.1/32	ISIS-L2	15	10	D	10.0.12.1	Serial1/0/0
10.0.3.3/32	ISIS-L1	15	10	D	10.0.234.3	GigabitEthernet0/0/0
10.0.4.4/32	ISIS-L1	15	10	D	10.0.234.4	GigabitEthernet0/0/0
10.0.5.5/32	ISIS-L2	15	20	D	10.0.12.1	Serial1/0/0
10.0.13.0/24	ISIS-L1	5	20	D	10.0.234.3	GigabitEthernet0/0/0
10.0.15.0/24	ISIS-L2	15	20	D	10.0.12.1	Serial1/0/0

ISIS routing table status : <Inactive>

Destinations : 0 Routes : 0

[R3]display isis route

Route information for ISIS(1)

ISIS(1) Level-1 Forwarding Table

IPv4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags

0.0.0.0/0	10	NULL			
10.0.4.4/32	10	NULL	GE0/0/0	10.0.234.4	A/-/L/-
10.0.12.0/24	20	NULL	GE0/0/0	10.0.234.2	A/-/L/-
10.0.13.0/24	10	NULL	GE0/0/1	Direct	D/-/L/-
10.0.234.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.2.2/32	10	NULL	GE0/0/0	10.0.234.2	A/-/L/-
10.0.3.3/32	0	NULL	Loop0	Direct	D/-/L/-

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,

U-Up/Down Bit Set

ISIS(1) Level-2 Forwarding Table

IPv4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags

10.0.4.4/32	20	NULL			
10.0.5.5/32	20	NULL	GE0/0/1	10.0.13.1	A/-/-/-
10.0.12.0/24	20	NULL			
10.0.13.0/24	10	NULL	GE0/0/1	Direct	D/-/L/-
10.0.234.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.15.0/24	20	NULL	GE0/0/1	10.0.13.1	A/-/-/-
10.0.1.1/32	10	NULL	GE0/0/1	10.0.13.1	A/-/-/-
10.0.2.2/32	10	NULL			
10.0.3.3/32	0	NULL	Loop0	Direct	D/-/L/-

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,

U-Up/Down Bit Set

[R3]display ip routing-table protocol isis

Route Flags: R - relay, D - download to fib

Public routing table : ISIS

Destinations : 6 Routes : 6

ISIS routing table status : <Active>

Destinations : 6 Routes : 6

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
------------------	-------	-----	------	-------	---------	-----------

10.0.1.1/32	ISIS-L2	15	10	D	10.0.13.1	GigabitEthernet0/0/1
10.0.2.2/32	ISIS-L1	15	10	D	10.0.234.2	GigabitEthernet0/0/0
10.0.4.4/32	ISIS-L1	15	10	D	10.0.234.4	GigabitEthernet0/0/0
10.0.5.5/32	ISIS-L2	15	20	D	10.0.13.1	GigabitEthernet0/0/1
10.0.12.0/24	ISIS-L1	15	20	D	10.0.234.2	GigabitEthernet0/0/0
10.0.15.0/24	ISIS-L2	15	20	D	10.0.13.1	GigabitEthernet0/0/1

ISIS routing table status : <Inactive>

Destinations : 0 Routes : 0

R4 is a Level-1 router and can only establish IS-IS neighbor relationships with Level-1 or Level-1-2 routers in the same area. By default, a Level-1 router cannot learn routing information of Level-2 routers and can only access external networks through default routes. The following command output shows that R4 has two default routes pointing to R2 and R3, and the two routes work in load balancing mode.

[R4]display isis route

Route information for ISIS(1)

ISIS(1) Level-1 Forwarding Table

IPv4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags

0.0.0.0/0	10	NULL	GE0/0/0	10.0.234.3	A/-/-/-
			GE0/0/0	10.0.234.2	

10.0.4.4/32	0	NULL	Loop0	Direct	D/-/L/-
10.0.12.0/24	20	NULL	GE0/0/0	10.0.234.2	A/-/-/-
10.0.13.0/24	20	NULL	GE0/0/0	10.0.234.3	A/-/-/-
10.0.234.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.2.2/32	10	NULL	GE0/0/0	10.0.234.2	A/-/-/-
10.0.3.3/3	10	NULL	GE0/0/0	10.0.234.3	A/-/-/-

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,

U-Up/Down Bit Set

[R4]display ip routing protocol isis

Route Flags: R - relay, D - download to fib

Public routing table : ISIS

Destinations : 5 Routes : 6

ISIS routing table status : <Active>

Destinations : 5 Routes : 6

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	ISIS-L1	15	10	D	10.0.234.3	GigabitEthernet0/0/0
	ISIS-L1	15	10	D	10.0.234.2	GigabitEthernet0/0/0
10.0.2.2/32	ISIS-L1	15	10	D	10.0.234.2	GigabitEthernet0/0/0
10.0.3.3/32	ISIS-L1	15	10	D	10.0.234.3	GigabitEthernet0/0/0
10.0.12.0/24	ISIS-L1	15	20	D	10.0.234.2	GigabitEthernet0/0/0
10.0.13.0/24	ISIS-L1	15	20	D	10.0.234.3	GigabitEthernet0/0/0

ISIS routing table status : <Inactive>

Destinations : 0 Routes : 0

Before configuring IS-IS to import external routes, check route learning.

[R5]display isis route

Route information for ISIS(1)

ISIS(1) Level-2 Forwarding Table

IPv4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags
10.0.4.4/32	30	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.5.5/32	0	NULL	Loop0	Direct	D/-/L/-
10.0.12.0/24	20	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.13.0/24	20	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.234.0/24	30	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.15.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.1.1/32	10	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.2.2/32	20	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.3.3/32	20	NULL	GE0/0/0	10.0.15.1	A/-/-/-

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,

U-Up/Down Bit Set

[R5]display ip routing-table protocol isis

Route Flags: R - relay, D - download to fib

Public routing table : ISIS

Destinations : 7

Routes : 7

ISIS routing table status : <Active>

Destinations : 7 Routes : 7

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.1.1/32	ISIS-L2	15	10	D	10.0.15.1	GigabitEthernet0/0/0
10.0.2.2/32	ISIS-L2	15	20	D	10.0.15.1	GigabitEthernet0/0/0
10.0.3.3/32	ISIS-L2	15	20	D	10.0.15.1	GigabitEthernet0/0/0
10.0.4.4/32	ISIS-L2	15	30	D	10.0.15.1	GigabitEthernet0/0/0
10.0.12.0/24	ISIS-L2	15	20	D	10.0.15.1	GigabitEthernet0/0/0
10.0.13.0/24	ISIS-L2	15	20	D	10.0.15.1	GigabitEthernet0/0/0
10.0.234.0/24	ISIS-L2	15	30	D	10.0.15.1	GigabitEthernet0/0/0

ISIS routing table status : <Inactive>

Destinations : 0 Routes : 0

Create new loopback interfaces on R5 and import direct routes into the Level-2 routing table in IS-IS process 1.

```
[R5]interface LoopBack 1
```

```
[R5-LoopBack1]ip address 192.168.1.1 24
```

```
[R5-LoopBack1]quit
```

```
[R5]interface LoopBack 2
```

```
[R5-LoopBack2]ip address 192.168.2.1 24
```

```
[R5-LoopBack2]quit
```

```
[R5]interface LoopBack 3
```

```
[R5-LoopBack3]ip address 192.168.3.1 24
```

```
[R5-LoopBack3]quit
```

```
[R5]isis
```

```
[R5-isis-1]import-route direct level-2
```

```
[R5-isis-1]quit
```

Check IS-IS routes of R5.

```
[R5]display isis route
```

Route information for ISIS(1)

ISIS(1) Level-2 Forwarding Table

IPv4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags

10.0.4.4/32	30	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.5.5/32	0	NULL	Loop0	Direct	D/-/L/-
10.0.12.0/24	20	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.13.0/24	20	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.234.0/24	30	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.15.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.1.1/32	10	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.2.2/32	20	NULL	GE0/0/0	10.0.15.1	A/-/-/-
10.0.3.3/32	20	NULL	GE0/0/0	10.0.15.1	A/-/-/-

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,

U-Up/Down Bit Set

ISIS(1) Level-2 Redistribute Table

Type	IPv4 Destination	IntCost	ExtCost	Tag
------	------------------	---------	---------	-----

D	192.168.1.0/24	0	0	
D	192.168.2.0/24	0	0	
D	192.168.3.0/24	0	0	

Type: D-Direct, I-ISIS, S-Static, O-OSPF, B-BGP, R-RIP, U-UNR

Check the IS-IS routing table of R4 again. No changes are found. This is because Level-2 routes are not leaked into Level-1 routers by default. R4 can access 192.168.1.0/24, 192.168.2.0/24, and 192.168.3.0/24 through default routes.

[R4]display ip routing-table protocol isis

Route Flags: R - relay, D - download to fib

Public routing table : ISIS

Destinations : 5 Routes : 6

ISIS routing table status : <Active>

Destinations : 5 Routes : 6

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
0.0.0.0/0	ISIS-L1	15	10	D	10.0.234.3	GigabitEthernet0/0/0
	ISIS-L1	15	10	D	10.0.234.2	GigabitEthernet0/0/0
10.0.2.2/32	ISIS-L1	15	10	D	10.0.234.2	GigabitEthernet0/0/0
10.0.3.3/32	ISIS-L1	15	10	D	10.0.234.3	GigabitEthernet0/0/0
10.0.12.0/24	ISIS-L1	15	20	D	10.0.234.2	GigabitEthernet0/0/0

10.0.13.0/24 ISIS-L1 15 20 D 10.0.234.3 GigabitEthernet0/0/0

ISIS routing table status : <Inactive>

Destinations : 0 Routes : 0

[R4]ping -c 1 192.168.1.1

PING 192.168.1.1: 56 data bytes, press CTRL_C to break

Reply from 192.168.1.1: bytes=56 Sequence=1 ttl=253 time=14 ms

--- 192.168.1.1 ping statistics ---

1 packet(s) transmitted

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 14/14/14 ms

[R4]ping -c 1 192.168.2.1

PING 192.168.2.1: 56 data bytes, press CTRL_C to break

Reply from 192.168.2.1: bytes=56 Sequence=1 ttl=253 time=13 ms

--- 192.168.2.1 ping statistics ---

1 packet(s) transmitted

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 13/13/13 ms

[R4]ping -c 1 192.168.3.1

PING 192.168.3.1: 56 data bytes, press CTRL_C to break

Reply from 192.168.3.1: bytes=56 Sequence=1 ttl=253 time=1 ms

--- 192.168.3.1 ping statistics ---

1 packet(s) transmitted

1 packet(s) received

0.00% packet loss

round-trip min/avg/max = 1/1/1 ms

Step 6 **Change the IS-IS interface cost.**

By default, the IS-IS interface cost is 10, which is not automatically calculated based on the bandwidth. For R1, traffic destined for R4 is load balanced between R2 and R3. R1 and R2 are connected using Serial interfaces, which have low bandwidth and are prone to bandwidth bottlenecks. Therefore, you can change the IS-IS interface cost to control route selection of R1.

Increase the outbound interface cost.

```
[R1]interface Serial 1/0/0
```

```
[R1-Serial1/0/0]isis cost 15
```

```
[R1-Serial1/0/0]quit
```

```
[R1]display isis route
```

Route information for ISIS(1)

ISIS(1) Level-2 Forwarding Table

IPv4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags

10.0.4.4/32	20	NULL	GE0/0/1	10.0.13.3	A/-/-/-

10.0.5.5/32	10	NULL	GE0/0/0	10.0.15.5	A/-/-/-
192.168.1.0/24	10	0	GE0/0/0	10.0.15.5	A/-/-/-
10.0.12.0/24	15	NULL	S1/0/0	Direct	D/-/L/-
192.168.2.0/24	10	0	GE0/0/0	10.0.15.5	A/-/-/-
10.0.13.0/24	10	NULL	GE0/0/1	Direct	D/-/L/-
192.168.3.0/24	10	0	GE0/0/0	10.0.15.5	A/-/-/-
10.0.234.0/24	20	NULL	GE0/0/1	10.0.13.3	A/-/-/-
10.0.15.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.1.1/32	0	NULL	Loop0	Direct	D/-/L/-
10.0.2.2/32	15	NULL	S1/0/0	10.0.12.2	A/-/-/-
10.0.3.3/32	10	NULL	GE0/0/1	10.0.13.3	A/-/-/-

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,
U-Up/Down Bit Set

The preceding command output shows that traffic from R1 to R4 is forwarded through Ethernet interfaces.

Step 7 **Configure IS-IS route leaking.**

Observe data forwarding on R4. R4 does not know Level-2 network information and forwards data to R2 and R3 for load balancing. If you do not want R4 to use the link between R2 and R1, configure route leaking to import Level-2 routes into Level-1. According to the longest match principle, R3 forwards packets destined for R5.

Before performing a tracer operation, enable interface unreachable response on R5. By default, interface unreachable response is disabled. If it is disabled, the last hop will expire.

Before interface unreachable response is enabled:

```
[R4]tracert 192.168.1.1
```

```
traceroute to 192.168.1.1(192.168.1.1), max hops: 30 ,packet length: 40,press CTRL_C to break
```

```
1 10.0.234.2 2 ms 10.0.234.3 2 ms 10.0.234.2 2 ms
```

```
2 10.0.13.1 11 ms 10.0.12.1 9 ms 10.0.13.1 11 ms
3 * * *
```

After interface unreachable response is enabled on R5:

```
[R5]icmp port-unreachable send
```

```
[R4]tracert 192.168.1.1
```

```
tracert to 192.168.1.1(192.168.1.1), max hops: 30 ,packet length: 40,press CTRL_C to break
```

```
1 10.0.234.2 2 ms 10.0.234.3 2 ms 10.0.234.2 2 ms
```

```
2 10.0.13.1 2 ms 10.0.12.1 9 ms 10.0.13.1 1 ms
```

```
3 192.168.1.1 8 ms 1 ms 9 ms
```

The preceding command output shows that tracert packets sent each time are load balanced between two next-hop addresses and then reach R5. You can enable route leaking on R3 to make R3 become the preferred next hop.

```
[R3]isis
```

```
[R3-isis-1]import-route isis level-2 into level-1
```

```
[R3-isis-1]quit
```

```
[R4]display isis route
```

```
Route information for ISIS(1)
```

```
-----
```

```
ISIS(1) Level-1 Forwarding Table
```

```
-----
```

IPV4 Destination	IntCost	ExtCost	ExitInterface	NextHop	Flags
------------------	---------	---------	---------------	---------	-------

0.0.0.0/0	10	NULL GE0/0/0		10.0.234.3	A/-/-/-
		GE0/0/0		10.0.234.2	
10.0.4.4/32	0	NULL	Loop0	Direct	D/-/L/-
10.0.5.5/32	30	NULL	GE0/0/0	10.0.234.3	A/-/-/U
192.168.1.0/24	10	20	GE0/0/0	10.0.234.3	A/-/-/U
10.0.12.0/24	20	NULL	GE0/0/0	10.0.234.2	A/-/-/-
192.168.2.0/24	10	20	GE0/0/0	10.0.234.3	A/-/-/U
10.0.13.0/24	20	NULL	GE0/0/0	10.0.234.3	A/-/-/-
192.168.3.0/24	10	20	GE0/0/0	10.0.234.3	A/-/-/U
10.0.234.0/24	10	NULL	GE0/0/0	Direct	D/-/L/-
10.0.15.0/24	30	NULL	GE0/0/0	10.0.234.3	A/-/-/U
10.0.1.1/32	20	NULL	GE0/0/0	10.0.234.3	A/-/-/U
10.0.2.2/32	10	NULL	GE0/0/0	10.0.234.2	A/-/-/-
10.0.3.3/32	10	NULL	GE0/0/0	10.0.234.3	A/-/-/-

Flags: D-Direct, A-Added to URT, L-Advertised in LSPs, S-IGP Shortcut,

U-Up/Down Bit Set

[R4]display ip routing-table protocol isis

Route Flags: R - relay, D - download to fib

Public routing table : ISIS

Destinations : 11 Routes : 12

ISIS routing table status : <Active>

Destinations : 11 Routes : 12

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
------------------	-------	-----	------	-------	---------	-----------

0.0.0.0/0	ISIS-L1	15	10	D	10.0.234.3	GigabitEthernet0/0/0
	ISIS-L1	15	10	D	10.0.234.2	GigabitEthernet0/0/0
10.0.1.1/32	ISIS-L1	15	20	D	10.0.234.3	GigabitEthernet0/0/0
10.0.2.2/32	ISIS-L1	15	10	D	10.0.234.2	GigabitEthernet0/0/0
10.0.3.3/32	ISIS-L1	15	10	D	10.0.234.3	GigabitEthernet0/0/0
10.0.5.5/32	ISIS-L1	15	30	D	10.0.234.3	GigabitEthernet0/0/0
10.0.12.0/24	ISIS-L1	15	20	D	10.0.234.2	GigabitEthernet0/0/0
10.0.13.0/24	ISIS-L1	15	20	D	10.0.234.3	GigabitEthernet0/0/0
10.0.15.0/24	ISIS-L1	15	30	D	10.0.234.3	GigabitEthernet0/0/0
192.168.1.0/24	ISIS-L1	15	94	D	10.0.234.3	GigabitEthernet0/0/0
192.168.2.0/24	ISIS-L1	15	94	D	10.0.234.3	GigabitEthernet0/0/0
192.168.3.0/24	ISIS-L1	15	94	D	10.0.234.3	GigabitEthernet0/0/0

ISIS routing table status : <Inactive>

Destinations : 0 Routes : 0

[R4]tracert 192.168.1.1

traceroute to 192.168.1.1(192.168.1.1), max hops: 30 ,packet length: 40,press CTRL_C to break

```

1 10.0.234.3 2 ms  1 ms  1 ms
2 10.0.13.1 2 ms  2 ms  2 ms
3 192.168.1.1 1 ms  1 ms  2 ms

```

The preceding tracert operation proves the impact of route leaking on IS-IS route selection.

----End